

Wary: Product Requirements Document

Project: Wary, Roman Urdu health triage agents that know when not to answer **Category:** Health Care **Event:** Pak Angels GenAI & Agentic AI Training, Cohort 11, Mid-Term Hackathon **Team lead:** Aila Nasir **Status:** Working prototype (Gradio), 4 October 2026

1. Overview

Wary is a multi-agent health triage prototype for Roman Urdu speakers. It takes a free-text complaint in Roman Urdu, Urdu or English, runs it through four agents, and returns either a cautious screening result, an abstention, or an emergency escalation. A doctor must sign off on every case. The design principle is that the system is allowed to say “I should not answer this”.

2. Problem

- Health chatbots often sound confident when they are wrong. An independent 2026 evaluation reported in Nature Medicine found that ChatGPT Health under-triaged more than half of the cases it was given (reported by The Guardian, 26 February 2026).
- Most tools are built for English. Many Pakistanis describe symptoms in Roman Urdu or mixed Urdu and English.
- Most symptom checkers give no audit trail, so a clinician cannot see why a decision was made.

3. Target users

User	Need
Patient or family member	Describe symptoms in their own language and get a safe next step
Doctor or clinic staff	A short handoff note and a record of how the decision was reached
Reviewer or researcher	A measurable safety layer, with audit trail and evaluation tools

4. Goals and non-goals

Goals

1. Understand Roman Urdu, Urdu and English complaints.
2. Never delay an emergency: red flags are detected by deterministic rules, not by an LLM.
3. Abstain when confidence is low, the top conditions are too close, or information is too thin.
4. Keep a human in the loop: every case is signed off by a doctor.
5. Make safety measurable (self-test and risk-coverage curve).

Non-goals

- Wary does not diagnose and does not prescribe.

- It is not a replacement for emergency services or a clinician.
- It is not validated on real patient data.

5. Solution and agent architecture

#	Agent	Responsibility
1	Intake	Converts Roman Urdu, Urdu or English text into structured symptoms (rule-based lexicon, optionally refined with Gemini). Extracts duration and language.
2	Specialist	Symptom classifier returns the top 3 conditions with raw scores. Chest X-ray and brain MRI modules (with Grad-CAM) act as optional evidence.
3	Safety Auditor	Deterministic checks: emergency red flags, low model confidence, close call between top 2 conditions, too little information, stricter threshold for age under 5 and 65 and above, and imaging flagged in the session. Outputs EMERGENCY, ABSTAIN, PRIORITY REVIEW or ROUTINE REVIEW.
4	Report	Writes a Roman Urdu patient message and an English doctor handoff note. Emergency and abstain messages are fixed templates, not LLM output.

After the agents, a doctor records Approved, Modified or Rejected with notes. The session can be exported as a PDF report, and each triage run saves a JSON audit trail.

6. Functional requirements

- **FR1:** Accept free text in Roman Urdu, Urdu script or English, with patient name, age and contact number.
- **FR2:** Show a decision badge, the reasons for the decision, and the agent pipeline with per-agent timing.

- **FR3:** Hide the predicted condition from the patient when the decision is EMERGENCY or ABSTAIN. The internal top 3 stays in the doctor note only.
- **FR4:** Provide a doctor sign-off step that updates the session record.
- **FR5:** Export a PDF session report and a JSON audit trail.
- **FR6:** Let the user run chest X-ray and brain MRI analysis; flagged imaging raises later cases to priority review.
- **FR7:** Provide a Safety Lab with an adjustable abstention threshold, a red-flag self-test, and a risk-coverage plot on the held-out test set.

7. Safety requirements

- **SR1:** Emergency rules cover chest pain, severe breathing difficulty, stroke signs, heavy bleeding, loss of consciousness, seizure, severe allergic reaction, poisoning, pregnancy with bleeding, and thoughts of self-harm.
- **SR2:** Simple negation (for example “no chest pain”) is handled, and when in doubt the system escalates rather than suppresses.
- **SR3:** Emergency messages tell the user to go to the nearest hospital or call Rescue 1122 where available. No unverified helpline numbers are shown.
- **SR4:** Model scores are shown as raw scores, never as calibrated probabilities.
- **SR5:** Every output carries a disclaimer: AI-assisted screening, not a diagnosis.

8. Evaluation plan

Check	Method	Current status
Red-flag detection	30 hand-written Roman Urdu, Urdu and English cases, including negations	Demo self-test built into the app; hand-written, not a clinical validation
Abstention trade-off	Risk-coverage curve: share of queries answered versus accuracy on the answered ones, on the held-out symptom test set	Built into the Safety Lab
Roman Urdu coverage	Planned: evaluate on a larger Roman Urdu health-query benchmark with an emergency category	Planned
Clinical review	Planned: review by a licensed doctor	Planned

9. Technology

Python, scikit-learn (TF-IDF and logistic regression symptom model), PyTorch (chest X-ray and brain MRI classifiers with Grad-CAM), Google Gemini (optional intake refinement and friendly explanations), Gradio (interface), fpdf2 (PDF reports), matplotlib. Prototype runs in Google Colab.

10. Limitations and ethics

- Prototype only; not validated on real patient data. Do not use for real medical decisions.
- The symptom model is trained on a public English symptom dataset, so Roman Urdu support depends on the translation lexicon, which is limited.
- Red-flag rules are keyword-based and can miss unusual phrasing.
- Scores come from models that have not been calibrated for deployment.
- Patient details are entered per session and not stored beyond the session export the user chooses to download.

11. Roadmap

1. Evaluate on a larger Roman Urdu benchmark, with an emergency category as ground truth.
2. Add a doctor-reviewed red-flag lexicon.
3. Calibrate model scores and tune thresholds per age group.
4. Deploy on Hugging Face Spaces.
5. Add voice input for low-literacy users.

12. Team

Role	Name
Team lead, ML and safety auditor	Aila Nasir
LLM and agent developer	To be added
Data and evaluation developer	To be added
UI, slides and demo	To be added